

#Naïve Bayes

— Bayes Theorem

$$\hookrightarrow P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

— Bayesian Classification

→ Statistical classifier

— Naïve Bayes Classifier

→ Supervised Machine-learning Algorithm (labels given)

→ Assumes features are statistically independent

— Model Types

① Gaussian (Continuous / Non-discrete)

② Multinomial

③ Bernoulli (Boolean)

— Naïve Bayes Applications

① Medical application

② Real time predictions

③ Text classification

<Naïve Bayes is an eager learner>

Binary Classification → Yes/No (Boolean)

Multiclass classification → Multiple categories

Bayes Theorem

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

Conditional Probability

$$P(H|X) = \frac{P(H \cap X)}{P(X)}$$

Derived from Bayes Theorem,

$$P(c_i|X) = \frac{P(X|c_i)P(c_i)}{P(X)}$$

Since $P(X)$ is constant,

$P(c_i|X) = P(X|c_i)P(c_i)$ needs to be maximized.

True for Naïve Bayes

$$P(Y|x_1, x_2, \dots, x_N) = P(x_1|Y)P(x_2|Y) \dots P(x_N|Y)P(Y)$$

Example

instances/features

Person	Covid (y/n)	Flu (Y/N)	Fever(y/N)
1	y	N	y
2	N	y	y
3	y	y	y
4	N	N	N
5	y	N	y
6	N	N	y
7	y	N	y
8	y	N	N
9	N	y	y
10	N	y	N

A sample dataset
→ class

For the given instance, $X = (\text{Covid}, \text{Flu})$ Calculate,

- (i) Prior Probability, $P(\text{Fever}=y)$, $P(\text{Fever}=N)$
- (ii) Calculate conditional probability for the given instance
- (iii) Posterior Probability $P(\text{Fever}=y | \text{Covid}, \text{Flu})$
 $P(\text{Fever}=N | \text{Covid}, \text{Flu})$

- (iv) Determine whether fever = yes or fever = No.

Prior Probability = without evidence
Posterior Probability = with evidence

Example

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A sample dataset

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 Posterior Probability = with evidence

(i) Prior Probability

$$P(\text{Fever} = \text{yes}) = 7/10$$

$$P(\text{Fever} = \text{No}) = 3/10$$

Fever = Y, Covid = Y

(ii)

	Y (7)	N (3)
Covid	4/7	2/3
Flu	3/7	2/3

Conditional Probability

(iii) Posterior Probability

$$P(\text{Fever} = Y | \text{Covid}, \text{Flu})$$

$$= P(F|Y) P(C|Y) P(Y)$$

$$= \frac{4}{7} \times \frac{3}{7} \times \frac{7}{10}$$

$$= \cancel{0.17} \quad 0.17$$

$$P(\text{Fever} = N | \text{Covid}, \text{Flu})$$

$$= P(F|N) P(C|N) P(N)$$

$$= \frac{2}{3} \times \frac{2}{3} \times \frac{3}{10} = 0.19$$

(iv) fever = No

Example

i) ~~$P(c=yes) = 9/14$~~

~~$P(c=no) = 5/14$~~

$P(c=yes) = 9/14$ $P(c=no) = 5/14$

ii)

≤ 30
Y = 2/9 N = 3/5

medium
Y = 4/9 N = 2/5

Student
Y = 6/9 N = 4/5

c
Y = 6/9 N = 2/5

iii) Posterior Probability

→ iv) Predict $c=yes$ or $c=no$.

#Genetic Algorithm

→ Inspired by Darwin's methodologies.

Evolution Algorithm:

- | | | |
|---|---------------|------------|
| { | Keywords : | |
| | → Gene | |
| | → Chromosomes | |
| | → Fitness | Fitness fn |
| | → Crossover | Cross Prob |
| | → Mutation | Mut prob |
| | | Offspring |
| | | Population |

Shayan | 6

